

IT2212 Data Wrangling Report

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Chapter 1: Operational Objectives

The objective of this workflow was to construct a reliable analytical dataset by consolidating 4 independently maintained customer databases from the North, South, East, and West branches of Nanyang Trading Company. Prior to analysis, the datasets contained structural inconsistencies, duplicate customer records, invalid values, and personally identifiable information that required preprocessing before meaningful business insights could be generated.

The workflow was designed to achieve 4 operational objectives:

- standardise the structure of all 4 regional datasets before consolidation;
- identify and correct data quality issues that could distort statistical analysis;
- anonymise sensitive customer information in compliance with the PDPA; and
- transform customer attributes into analytical features suitable for demographic and behavioural analysis.

The completed workflow produces a clean, consistent, and analysis-ready dataset for customer spending analysis.

1.1 Workflow Architecture

The workflow consists of 5 processing stages performed sequentially.

1. Data Mashup
2. Data Cleaning
3. Data Anonymisation
4. Data Transformation
5. Data Visualisation

Each stage performs a specific function and prepares the dataset for the subsequent stage, ensuring that analytical results are generated only from validated and consistent customer records.

Chapter 2: Data Mashup

2.1 Standardising the Dataset Structure

The data mashup stage began by importing the 4 regional customer datasets (North, South, East, and West) into KNIME using 4 Excel Reader nodes. Importing the datasets separately allowed each regional dataset to undergo individual preprocessing before consolidation, ensuring that structural inconsistencies could be resolved independently without affecting the other datasets.

Following the import, it was observed that the customers' name structure differed across the 4 datasets. The East and West datasets stored customer names in separate First Name and Last Name columns, whereas the North and South datasets stored both names within a single Name column. If the datasets were concatenated immediately, identical customer information would exist under different schemas, resulting in inconsistent attribute definitions.

To resolve this inconsistency, a String Manipulator node was first applied to the North and South Datasets, which removed any leading and trailing whitespace from the Name column. Performing this standardisation before splitting ensured that every customer name followed a consistent format.

The cleaned Name column was then passed into a Cell Splitter node, which separated the combined customer name into 2 independent attributes. As a result, the North and South datasets adopted the same name structure as the East and West datasets, allowing all 4 datasets to follow a common schema.

After splitting, the generated columns were labelled Name_Arr[0] and Name_Arr[1]. A Column Renamer node was therefore used to rename these attributes as First Name and Last Name respectively. Standardising the column names before merging was essential as the Concatenate node matches columns based on their names. If inconsistent names were retained, KNIME would interpret them as different attributes, resulting in duplicated columns and missing values within the final dataset.

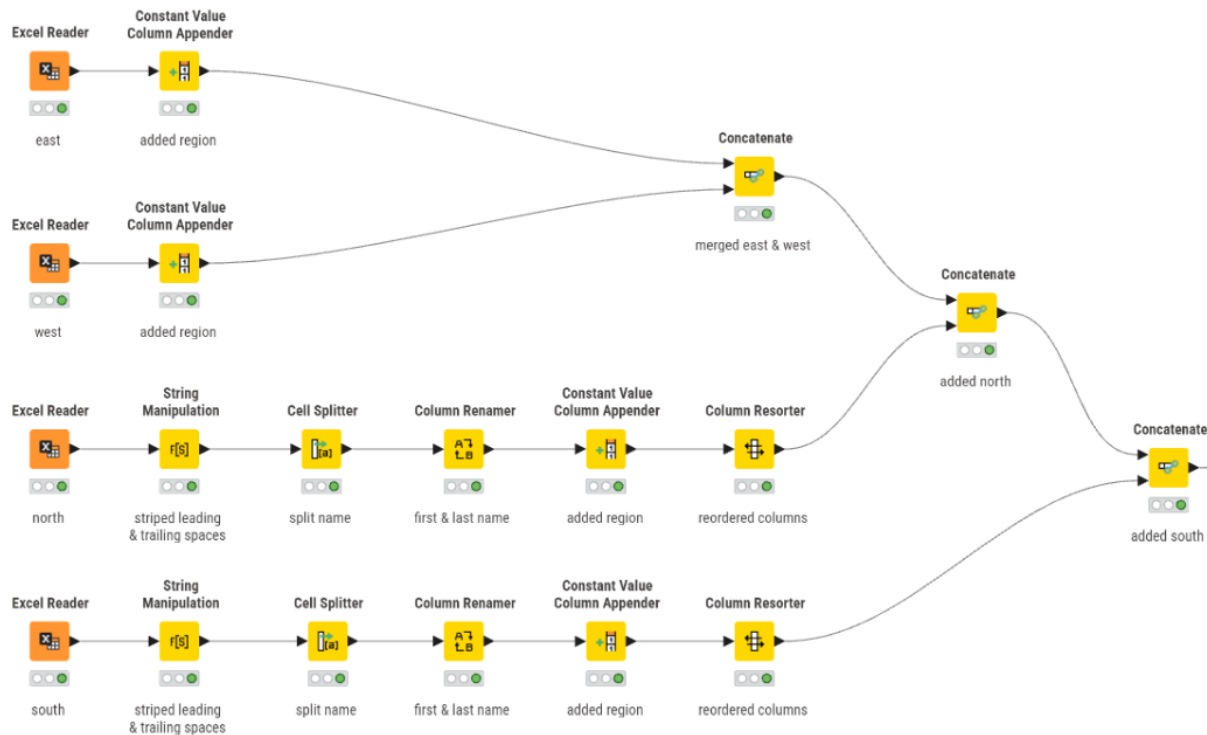
2.2 Appending Regional Information

Once the structure of all 4 datasets had been standardised, a Constant Value Column Appender node was applied to each regional dataset to create a region column. Each dataset was assigned a fixed value corresponding to its original branch (North, South, East, or West).

Appending the Region column before consolidation ensured that every customer record retained its geographical origin after the datasets were merged, allowing regional spending analysis to be performed accurately.

2.3 Preparing the Datasets for Consolidation

A Column Resorter node was applied to each dataset to arrange the columns into a consistent sequence before consolidation, which ensures that every regional dataset follows the same schema.



Chapter 3: Data Cleaning & Anonymisation

3.1 Removing Duplicate Customer Records

Following the data mashup stage, duplicate customer records were identified as the first data quality issue to be addressed. Since the SSN column uniquely identifies each customer, duplicate detection was performed using the SSN attribute rather than customer names. This avoids incorrectly identifying different customers with identical names as duplicates, while allowing repeated records belonging to the same customer to be identified correctly.

A Duplicate Row Filter node was configured to identify repeated SSN values within the consolidated dataset. The duplicate check affected 11 records, of which 6 were identified as duplicate records. The repeated records were reviewed and contained the same SSN and matching values across the other customer attributes, confirming that they represented repeated customer records rather than different customers who happened to share the same identifier.

☐	#	RowID	SSN T: String	First Name T: String	Last Name T: String	Birthdate Date	Cred Card... T: String	Income Number (L...	Purchases Number (L...	Amount S... Number (L...	Region T: String	Duplic... T: String
☐	376	Row37	411-80-3744	Levi	Wolfe	1974-09-06	0	84200	3	1410	West	chosen
☐	483	Row48	283-42-4994	Jeanne	Mendoza	1966-05-22	0	133200	3	1460	West	chosen
☐	761	Row76	638-19-2849	Alexander	Barnett	1930-07-31	0	121100	5	2540	North	chosen
☐	770	Row76	166-84-2698	Ricardo	Marsh	1966-10-30	0	91000	6	2910	North	chosen
☐	1164	Row11	680-00-1375	Kenny	Lindsey	1948-09-28	1	139500	4	2050	South	chosen
☐	544	Row54	283-42-4994	Jeanne	Mendoza	1966-05-22	0	133200	3	1460	West	duplicate
☐	736	Row73	411-80-3744	Levi	Wolfe	1974-09-06	0	84200	3	1410	West	duplicate
☐	790	Row78	638-19-2849	Alexander	Barnett	1930-07-31	0	121100	5	2540	North	duplicate
☐	925	Row92	638-19-2849	Alexander	Barnett	1930-07-31	0	121100	5	2540	North	duplicate
☐	1159	Row11	166-84-2698	Ricardo	Marsh	1966-10-30	0	91000	6	2910	North	duplicate
☐	1169	Row11	680-00-1375	Kenny	Lindsey	1948-09-28	1	139500	4	2050	South	duplicate

Figure 2. Records affected by duplicate SSN detection, including 6 duplicate customer records.

The duplicate records were then subsequently removed using another Duplicate Row Filter node, retaining only the first complete record for each customer. This was necessary because retaining repeated customer records would cause individual customers to be counted more than once in subsequent calculations. Removing the duplicates ensured that each customer contributed only one observation to the later analysis and prevented the same customer's spending behaviour from being disproportionately represented.

3.2 Correcting Invalid Birthdates

The Birthdate attribute was then examined for implausible values. A Date&Time Part Extractor node was first used to extract the year component into a separate Birth Year column. Several records contained birth years later than 2008, indicating customers below 18 years old. As an analytical assumption, customers below 18 were considered unsuitable for the intended adult customer spending analysis. These values were therefore treated as invalid birth-year values for data preparation purposes.

☐	#	RowID	SSN T: String	First Name T: String	Last Name T: String	Birthdate T: Date	Cred Card... T: String	Income N: Number (L...)	Purchases N: Number (L...)	Amount S... N: Number (L...)	Region T: String	Birth ... N: Number (L...)
☐	820	Row82	237-88-3817	Cameron	Fletcher	2029-08-11	0	63800	8	3960	North	2029
☐	123	Row12	618-84-1169	Garrett	Kelly	2028-10-19	1	113600	1	470	East	2028
☐	302	Row30	806-70-0226	Lynn	Freeman	2028-10-14	0	40600	4	1960	East	2028
☐	326	Row32	380-84-2860	Jeremiah	Parsons	2028-10-17	0	91800	2	980	East	2028
☐	619	Row61	329-51-3208	Clara	Byrd	2028-03-21	1	35400	6	3000	West	2028
☐	1249	Row12	133-99-5496	Lynn	Adkins	2028-05-13	0	142500	2	1000	South	2028
☐	1272	Row12	968-16-0774	Regina	Gomez	2028-09-29	0	68400	2	1100	South	2028
☐	155	Row15	776-44-8345	Jake	Brady	2027-04-16	1	59300	2	1030	East	2027
☐	373	Row37	964-27-4755	Elbert	Maxwell	2027-01-29	0	26700	1	450	West	2027
☐	1046	Row10	808-29-7482	Felipe	Clarke	2027-02-28	0	90000	5	2580	North	2027
☐	1358	Row13	376-25-7809	Sergio	Leonard	2027-11-02	1	9999	2	1070	South	2027
☐	642	Row64	594-47-1955	Lonnie	Gross	2025-08-10	1	128300	3	1510	West	2025

Figure 3. Records containing birth years later than 2008 during the validation of the Birthdate attribute.

During this validation, I also found 18 records with a Birthdate of 17 May 1927. Although the repeated date resulted in an unusually high number of customers sharing the same birthday, the Birthdate itself was valid and did not violate the age criteria used for the analysis. The records were therefore retained, as there was insufficient evidence to conclude that the repeated birthdate was erroneous or that the Birthdate values should be replaced solely because the records shared the same date. In contrast, birth years after 2008 were treated as unsuitable for the intended adult-customer analysis, while birthdates after 31 December 2026 were treated as invalid because they represented future birthdates.

☐	134	Row13	364-94-9180	Ira	Stevens	1927-05-17	0	116500	2	1040	East	1927
☐	172	Row17	212-01-7062	Betty	Stewart	1927-05-17	0	147900	5	2450	East	1927
☐	183	Row18	142-06-2339	Wilbur	Yates	1927-05-17	1	38200	1	510	East	1927
☐	200	Row19	183-25-0406	Alma	Williamson	1927-05-17	0	51600	0	0	East	1927
☐	332	Row33	678-19-0332	Belinda	Mendoza	1927-05-17	0	83900	5	2710	East	1927
☐	419	Row41	205-84-3572	Leah	Sanchez	1927-05-17	0	50500	1	490	West	1927
☐	499	Row49	619-94-0553	Ellen	Garza	1927-05-17	0	62300	2	930	West	1927
☐	501	Row50	365-18-7407	Elaine	Moody	1927-05-17	0	155400	4	1900	West	1927
☐	510	Row50	085-32-5438	Rosa	Gilbert	1927-05-17	0	103700	1	480	West	1927
☐	593	Row59	338-58-7652	Pat	Schneider	1927-05-17	1	47500	2	1020	West	1927
☐	668	Row66	183-40-5102	Gregg	Haynes	1927-05-17	0	44800	4	1940	West	1927
☐	894	Row89	279-23-7773	Roxanne	Baldwin	1927-05-17	0	120300	4	2100	North	1927
☐	931	Row93	626-04-1182	Harry	Sanders	1927-05-17	1	75900	3	1540	North	1927
☐	1038	Row10	715-28-2884	Pedro	Bennett	1927-05-17	0	82400	4	1850	North	1927
☐	1317	Row13	086-39-4715	Warren	Leonard	1927-05-17	0	155300	5	2480	South	1927
☐	1345	Row13	891-12-9133	Sandy	Boyd	1927-05-17	0	77300	4	1980	South	1927
☐	1406	Row14	110-67-7322	Pamela	Barton	1927-05-17	0	138500	3	1400	South	1927
☐	1426	Row14	602-63-2343	Sam	Parker	1927-05-17	1	67800	3	1520	South	1927

Figure 4. The 18 records with the birthdate of 17 May 1927.

Rather than removing these customer records, the invalid birth years were corrected using median imputation. This approach was selected as Birthdate was not directly used as the primary analytical variable. Removing the entire customer record would unnecessarily discard the customer's valid purchasing behaviour, while replacing only the invalid birth year allowed the rest of the valid customer information to be retained for subsequent analysis.

A Column Expressions node was then used to identify Birth Year values greater than 2008 and convert them to null values while retaining all valid birth years unchanged, which ensured that the invalid birth years were excluded from the calculation of the replacement value. A Statistics node was then used to calculate the median from the remaining valid Birth Year values across the entire dataset, producing a median birth year of 1951. This median was used to apply a consistent correction across all regions.

Median imputation was preferred over mean imputation as the median is less sensitive to unusually early or late birth years. This made it a more robust representative value for correcting the invalid records without allowing extreme ages to disproportionately influence the replacement value. The Missing Value node was then used to replace the null Birth Year values with the calculated median of 1951, while valid Birth Year values remained unchanged.

To apply the corrected year to the original Birthdate values, the Birthdate column was converted from its date format into a string using a Date&Time to String node. A String Replacer node was then used to replace the invalid year component with 1951 while preserving the original month and day components. The corrected values were subsequently converted back into the appropriate date format using a String to Date&Time node. Although the Age column could have been calculated directly using a simplified formula based on Birth Year, correcting the original Birthdate field first ensured that the underlying date attribute was valid before it was used for further analysis. This also preserved the existing day and month information instead of discarding the complete age and calculating age using only the year component.

The corrected Birthdate values were subsequently used to derive customer age. A Date&Time Difference node calculated the difference between each Birthdate and a fixed reference date of 31 December 2026, with the output granularity set to years. This operation produced the new Age column. A fixed reference date was selected to ensure that the age calculation was consistent and reproducible for all records, rather than changing depending on the date on which the workflow was executed.

- used date and time part extractor node to get birth year from birthdate column
- used column expressions node to null invalid years (e.g. birth year > 2008)
- used missing value node to replace the nulled years with median year
- changed birthdate to string format temporarily to use string replacer node to replace invalid years with the median year
- changed birthdate back to its correct format: date format using the string to date & time node
- calculated customers' age using the date and time difference node by using the birthdate column

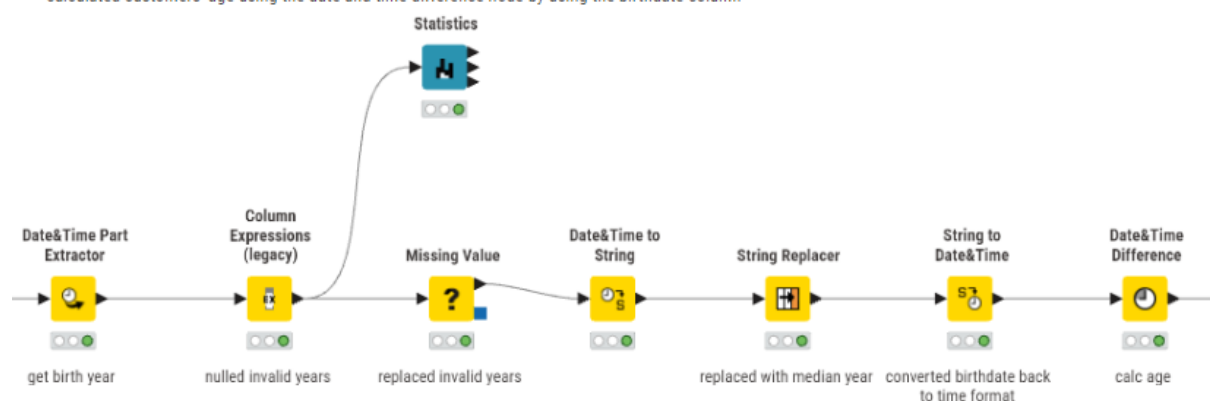


Figure 5. KNIME workflow for correcting invalid birthdates.

3.3 Removing Invalid Transaction Records

The transactional attributes were subsequently examined for values that could compromise the accuracy of the spending analysis. Several records contained the

placeholder value 9999 in the Purchases and Amount Spent columns. These values were treated as invalid as 9999 was not a realistic representation of the number of purchases or the amount spent by a customer, and was instead identified as a placeholder or data-entry error.

A Rule Engine node was used to identify records containing the invalid transaction placeholders and assign them an “Remove” status, while valid records were assigned a “Keep” status. The resulting status was appended as an Invalid Record column, which was then used by a Rule-based Row Filter to remove the affected records from the analytical dataset. After the 19 invalid records were removed, the temporary Invalid Record column was removed as it was not needed for subsequent analysis.

The decision to remove these records rather than replace them with the median was based on both the nature and proportion of the errors. The affected records represented less than 5% of the original dataset ($19/1500 \approx 1.2\%$), meaning that their removal had a limited effect on the overall sample size. Additionally, Purchases and Amount Spent are behavioural variables that directly form the basis of the spending analysis. Replacing an invalid placeholder such as 9999 with the median would create artificial purchasing behaviour that was not observed in the original data. Removing the small number of clearly invalid records was therefore considered more appropriate than introducing estimated transaction values.

The Income attribute was also reviewed for unusual values, including a value of 71 and several values of 9999. However, rows were not removed solely because an income value appeared unusually low or high. The unusual Income values could not be conclusively classified as invalid based only on their numerical magnitude. Removing records solely on their income values could therefore result in the unnecessary loss of otherwise usable customer records.

Median imputation was also not applied to the unusual Income values because there was insufficient evidence to confirm that they were erroneous. Replacing them with the median could overwrite potentially valid information and introduce estimated values into the dataset. For example, one record contained an Income value of 71, but was removed because both Purchases and Amount Spent contained the invalid placeholder value 9999. The removal was therefore based on the confirmed transactional errors, rather than the unusual Income value.

Income Number (L...	Purchases Number (L...	Amount S... Number (L...	Region String	Birth Year Number (...)	Age Number (...)	Invalid... String
64400	9999	9999	East	1,934	92	Remove
29900	9999	9999	East	1,971	55	Remove
104500	9999	9999	East	1,967	59	Remove
141200	9999	9999	East	1,952	74	Remove
93000	9999	9999	East	1,951	75	Remove
59300	9999	9999	East	1,930	96	Remove

Figure 6. 6 out of the 19 records identified for removal due to invalid Purchases or Amount Spent values.

A second logical validation was then performed on the relationship between Purchases and Amount Spent. A Rule Engine was used to identify records where Purchases = 0 while Amount Spent was greater than 0, and these records were assigned a “Remove” status, while all other records were assigned a “Keep” status, with the resulting status stored in a Row Status column. A Rule-based Row Filter then used this status to remove the 6 affected records from the dataset. After the 6 records were removed, the temporary Row Status column was also removed as it was no longer required for subsequent analysis.

This was considered logically inconsistent as a customer should not have a positive recorded expenditure without any recorded purchases. The 6 records also represented less than 5% ($6/1481 \approx 0.4\%$) of the dataset. They were therefore removed rather than having their values artificially imputed, as replacing either value could create an artificial relationship between the 2 transactional attributes while the small number of affected records meant that their removal had a limited impact on the overall dataset.

Purchases Number (L...	Amount S... Number (L...	Region String	Birth Year Number (...)	Age Number (...)	Row ... String
2	0	East	1,953	73	Remove
4	0	East	1,957	69	Remove
3	0	West	1,930	96	Remove
2	0	North	1,959	67	Remove
1	0	South	1,943	83	Remove
4	0	South	1,976	50	Remove

Figure 7. Records where Purchases = 0 while Amount Spent was greater than 0.

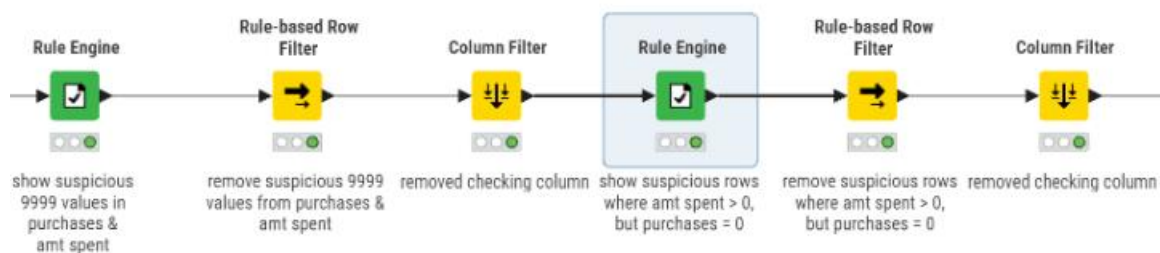


Figure 8. KNIME workflow for removing invalid transaction records.

Chapter 4: Data Transformation

4.1 Creating Age Groups

After the data cleaning and validation steps were completed, the cleaned Age column was transformed into broader Age Group categories using a Numeric Binner node. The following age intervals were configured:

- 49-59
- 60-69
- 70-79
- 80-89
- 90+

Grouping customers into age ranges was appropriate because age is a demographic characteristic that can reflect different life stages and potentially different spending behaviours. Using broader age categories allows the analysis to identify whether spending patterns may vary between different age segments, rather than focusing on minor differences between individual ages.

4.2 Creating Income Groups

The Income attribute was also transformed into broader income categories using a Numeric Binner node, producing an Income Group column. The following income intervals were configured:

- Low: below 50000
- Middle: 50000 to below 100000
- Upper Middle: 100000 to 150000
- High: above 150000

Grouping customers into income bands was appropriate because income represents the customer's financial capacity. The defined bands allow the analysis to assess whether average spending differs between customers with lower and higher income levels, while also making it possible to identify whether spending increases consistently as income rises or whether certain income segments behave differently.

4.3 Anonymising Customer Information

The customer-identifying attributes were then transformed to reduce the exposure of personally identifiable information in the final dataset. The First Name column was anonymised by replacing the original first names with asterisks (*) using the String Replacer node. Only the First Name column was masked while the Last Name remained unmasked because surnames are generally less effective as unique identifiers, as

multiple customers may share the same last name. Hence, masking the First Name therefore reduced the ability to directly identify individual customers.

The Full Name field was then formed using the masked First Name together with the unmasked Last Name. This ensured that the original First Name was not directly exposed while retaining the required name structure.

The SSN column was anonymised using hashing through the Anonymisation node. This transformed the original SSN values into anonymous hash values while maintaining a consistent identifier for each customer. As a result, records could still be distinguished and linked during analysis without exposing the original SSNs.

An Excel Writer node was connected to the Anonymisation node to save the corresponding mapping between the original SSN values and their anonymised hash values. This lookup table was kept separately from the analytical dataset and provides a controlled reference for authorised reversal, auditing or verification if required. The analytical dataset therefore uses the anonymised SSN values, while the original-to-anonymous mapping is maintained separately rather than being included directly in the analysis dataset.

For the Cred Card User column, the original binary coding of 1 and 0 was retained. A binary numerical representation is appropriate for this attribute because it clearly represents 2 mutually exclusive states and can be directly used for grouping and comparison during analysis. Retaining the existing coding also avoids introducing an unnecessary additional transformation when the underlying values already provide a consistent distinction between the 2 customer groups.

4.4 Removing Unnecessary Columns and Preparing the Final Dataset

After the aforementioned steps had been completed, 4 columns were removed using a Column Filter node: Birthdate, Birth Year, Age, Income. These columns were no longer required in the final analytical dataset. Birthdate and Birth Year contained detailed date information that had already been used to derive Age, while Age had already been transformed into the broader Age Group attribute. Similarly, the raw Income value had already been transformed into Income Group. Removing these 4 columns reduced unnecessary personal and intermediate information while retaining the transformed attributes required for analysis.

After the 4 unnecessary columns had been removed, a Column Resorter node was used to arrange the remaining attributes into a logical order. The columns were organised as SSN, Full Name, Cred Card User, Age Group, Income Group, Region, Purchases, and Amount Spent. This arrangement groups the anonymised customer information and demographic attributes before the regional and transactional variables, creating a clearer structure for the final dataset and subsequent visual analysis.

The final dataset was then exported using an Excel Writer node. This created a separate Excel file containing the cleaned, transformed, and anonymised dataset, allowing the final data to be preserved for submission and subsequent analysis. Exporting the dataset at this stage ensured that the saved file reflected the completed data preparation process, including the final column structure and the removal of unnecessary attributes.

Hence, the final dataset consists of 1475 rows and 8 columns.

- used numeric binner node to bin age and income - resulting in age and income groups
- masked first name using string replacer node to anonymize name
- combined masked first name with last name using column combiner node
- anonymized ssn by hashing it
- used excel writer node to save a secure lookup table matching original SSN to their new anonymous hashes so the data can be reversed if needed
- filtered out unneeded columns like birthdate, birth year, income, and age using column filter node
- resorted columns for easy reading using column resorter node

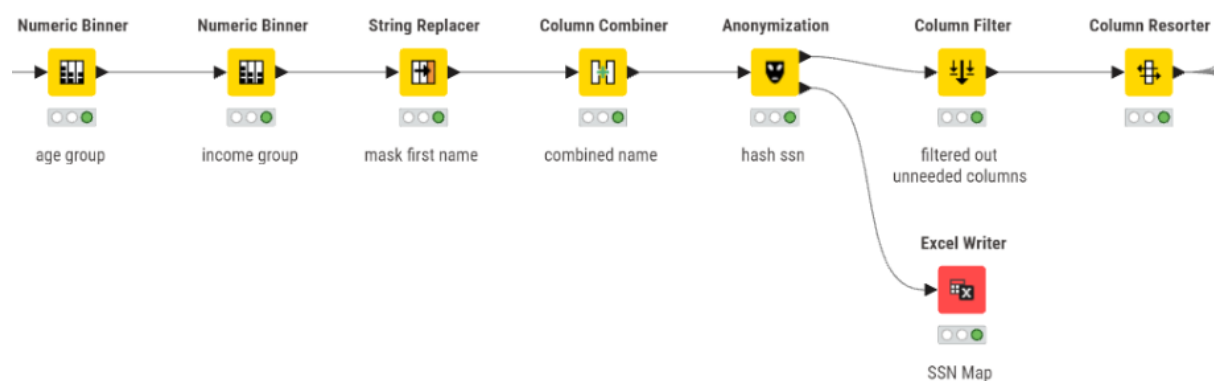


Figure 9. KNIME workflow for data transformation.

Rows: 1475 | Columns: 8

#	RowID	SSN	Full Name	Cred Card User	Age Group	Income Group	Region	Purchases	Amount Spent
		T: String	T: String	T: String	T: String	T: String	T: String	N: Number (Integer)	N: Number (Integer)

Figure 10. Final Dataset Structure.

Chapter 5: Data Visualisation

The final dataset was analysed using 6 visualisations to examine customer spending patterns across regional, demographic, and transactional attributes.

5.1 Average Amount Spent by Region and Credit Card User

Before creating the grouped bar chart, a GroupBy node was used to calculate the average Amount Spent for each combination of Region and Cred Card User. A Pivoting node then restructured the results to allow direct comparison between credit card users (coded 1), and non-credit card users (coded 0).

The grouped bar chart compares average spending across the 4 regions for credit card users (coded 1), and non-credit card users (coded 0).

The most noticeable difference occurred in West, where customers coded 1 had around a \$238 higher average spending level than customers coded 0. In contrast, the 2 groups showed very similar spending levels in East, with customers coded 1 recording a \$20 lower average spending than customers coded 0.

This suggests that the relationship between credit card user status and spending varies across regions. Although customers coded 1 recorded a higher spending in total (\$5974 vs \$5780), the pattern was not consistent enough to suggest that credit card user status alone explains spending behaviour. Instead, the regional differences indicate that other factors may influence the relationship between these 2 variables.

The grouped bar chart was used as it allows the 2 credit card user categories to be compared directly within each region, making both regional differences and differences between the 2 customer groups visible.

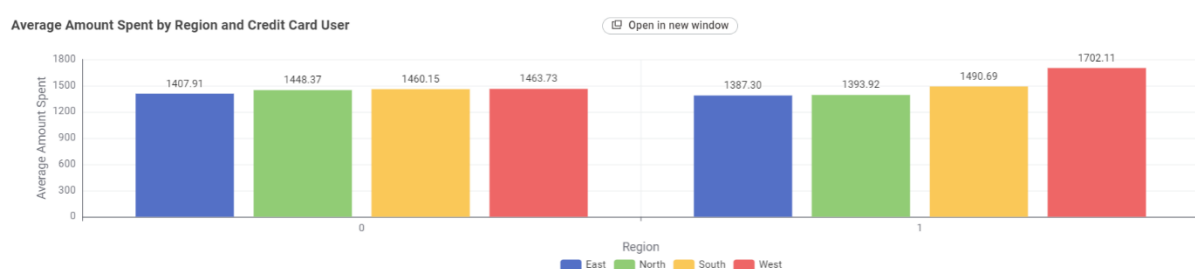


Figure 11. Grouped Bar Chart of Average Amount Spent by Region and Credit Card User.

5.2 Average Amount Spent by Region and Age Group

The heatmap compares average spending across 5 age groups and 4 regions. The results show that spending varies considerably between certain age-group and region combinations rather than following a consistent pattern based on age alone.

The highest spending combination was observed among customers aged 80-89 in West, while the lowest was observed among customers aged 60-69 in East. The difference

between these combinations was approximately \$400, indicating a notable variation in average spending.

The same age group also produced different spending patterns across regions. For example, the 60-69 age group recorded substantially different average spending levels between East and West (\$1236 for East, \$1590 for West), demonstrating that regional differences can be significant even when comparing customers within the same age category.

These results suggest that age alone does not provide a consistent explanation for customer spending. Older age groups do not consistently spend more across all regions, and the same age group can produce significantly different average spending levels depending on its region.

The heatmap was appropriate as it allows Region and Age Group to be examined simultaneously, making high and low-spending combinations easier to identify than analysing either variable independently.

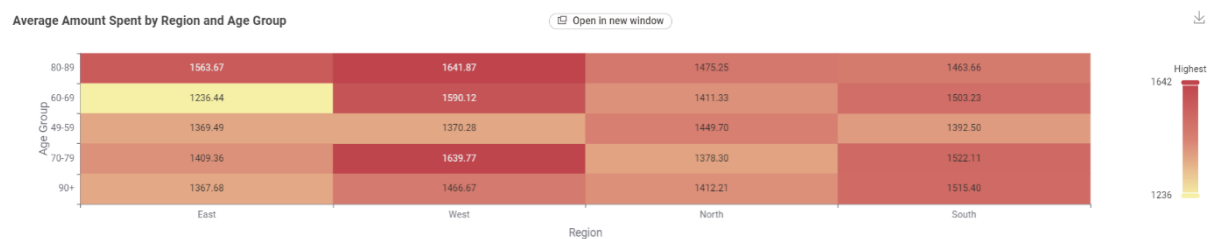


Figure 12. Heatmap showing Average Amount Spent by Region and Age Group.

5.3 Average Amount Spent by Age Group and Income Group

The heatmap examines average spending across combinations of age groups and income groups. The results show that the relationship between income and spending is not consistent across different age groups.

The highest spending combination was customers aged 60-69 in the “High” income group (\$1640), while the lowest was customers aged 49-59 also in the “High” income group (\$1099). The “High” income group also did not consistently record the highest spending across age groups, with the “Middle” income group exceeding it in some cases. This suggests that higher income does not automatically correspond to higher spending in this dataset.

Hence, age and income should not be interpreted independently when examining customer spending. The spending behaviour associated with an income group can differ depending on the customer’s age group.

The heatmap was used as it shows the combined relationship between 2 categorical variables and allows specific age-income combinations with unusually high or low spending to be identified.

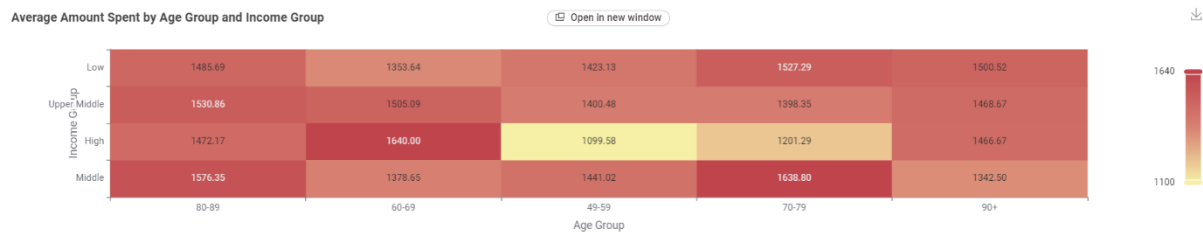


Figure 13. Heatmap showing Average Amount Spent by Age Group and Income Group.

5.4 Average Amount Spent by Income Group

The bar chart compares average spending across the 4 income groups. The results show relatively small differences between the overall average spending levels of the same categories.

The “Middle” income group recorded the highest average spending, while the “High” income group recorded the lowest. The difference between the highest and lowest groups was approximately \$130, indicating that the overall income-group averages were relatively close to one another.

The means do not rise steadily as Income Group increases. In particular, the “High” income group recorded the lowest average spending despite representing customers with the highest income classification. Therefore, income group alone is a weak standalone explanation of spending in this dataset.

This result is important as it demonstrates that higher financial capacity does not necessarily correspond to higher recorded spending. However, the overall bar chart does not show how income behaves differently across regions or age groups, which is why the heatmaps provide additional insight.

The bar chart was appropriate because it provides a direct comparison of average spending across the 4 income groups and gives an overall summary of income-based spending patterns.

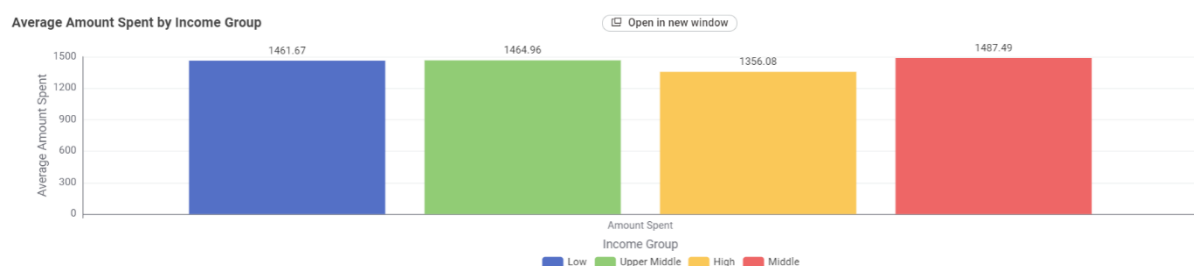


Figure 14. Bar Chart showing Average Amount Spent by Income Group.

5.5 Purchases vs Amount Spent

The scatter plot examines the relationship between Purchases and Amount Spent. Unlike the demographic comparisons, this visualisation shows a clear positive relationship between the 2 variables.

As the number of purchases increases, the general level of Amount Spent also increases. Customers with fewer purchases generally recorded lower spending, while customers with higher purchase counts generally recorded higher spending values. The upward pattern becomes particularly clear at the higher purchase counts.

Despite the overall positive relationship, some anomalies are visible in the scatter plot, where customers with a higher number of purchases can still have lower spending levels compared to customers with a lower number of purchases.

This suggests that Purchases is a stronger descriptive indicator of Amount Spent than the demographic segmentation variables examined in the previous visualisations. However, the relationship should still be interpreted as an association rather than evidence that a higher number of purchases directly causes higher spending.

The scatter plot was appropriate because both variables are numerical. Each point represents an individual customer, allowing the overall relationship and variation between customers to be observed simultaneously.



Figure 15: Scatter Plot showing Purchases vs Amount Spent.

5.6 Average Amount Spent by Region and Income Group

The final heatmap examines average spending across combinations of Region and Income Group. The results reveal substantially different spending patterns between particular regional and income combinations.

The highest average spending combination was observed among the “High” income group in West, while the lowest was observed among the “High” income group in South. The difference between these 2 combinations was over \$400, despite both groups belonging to the same income category.

The heatmap also shows that the same income group can produce different spending patterns depending on region. For example, the “High” income group recorded relatively high spending in West but relatively low spending in South. This indicates that income alone does not determine the spending behaviour observed within each region.

The heatmap therefore provides stronger evidence that region may influence the observed relationship between income and spending. Income group alone showed relatively small overall differences in average spending, but combining Income Group with Region revealed much larger variations between specific segments.

The heatmap was appropriate because it allows 2 categorical dimensions to be analysed simultaneously, making specific combinations with higher or lower average spending easier to identify.

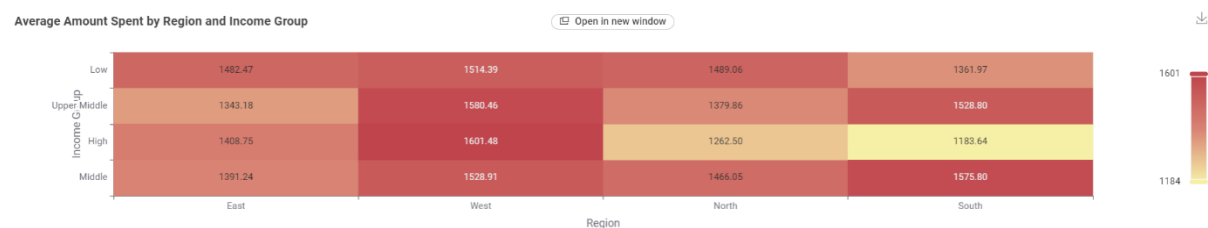


Figure 16. Heatmap showing Average Amount Spent by Region and Income Group.

5.7 Overall Findings

The 6 visualisations provide several key insights into customer spending behaviour.

The clearest relationship was observed between Purchases and Amount Spent. Customers with more purchases generally recorded higher total spending, suggesting that transaction frequency is a stronger descriptive indicator of total spending than the demographic variables examined in the other visualisations.

The results for Income Group were less straightforward. The overall income-group comparison showed relatively small differences in average spending, and the means did not increase consistently as income increased. The “High” income group even recorded the lowest overall average spending of \$1356. This suggests that income group alone is a weak standalone explanation of spending behaviour.

However, the heatmaps revealed that demographic variables can provide more useful insight when analysed together. The relationship between income and spending differed across regions and age groups, with the same income group recording substantially different average spending behaviour.

Similarly, age did not show a simple relationship with spending. Older customers did not consistently spend more than younger customers across all regions or income groups. Instead, spending patterns varied depending on the combination of age group and another categorical variable.

Overall, the analysis suggests that Purchases is the strongest descriptive indicator of Amount Spent, while Region, Age Group, and Income Group are more useful for identifying specific customer segments and combinations of characteristics. The findings also demonstrate the value of multi-dimensional analysis: income group alone showed relatively limited variation, but combining it with Region or Age Group revealed more substantial differences between specific customer segments.

These findings should be interpreted as descriptive relationships rather than causal conclusions. The visualisations identify patterns and differences within the processed

dataset, but they do not establish that a particular demographic characteristic directly causes customers to spend more or less.